

A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with faint, lighter blue diagonal stripes.

Cryptography



What is cryptography

- Cryptography is a way to hide information that only allow a particular person or people to have access to information



Modern application of cryptography

- Secure web browsing
 - Secure Socket Layer
 - Transport Layer Security
- End-to-end encrypted messaging (WhatsApp)
- Cryptocurrency (blockchain)
- Financial transactions
- Securing data at rest in cloud storage or on mobile devices



How does cryptography work

- Cryptography is a way of turning human readable message (plaintext) into a string of text that have no meaning (ciphertext)



Secure data at rest (Data storage)

Hashing and salting

Hash function is a encryption algorithm that perform hashing, turning human readable message into meaningless

Salting is a technique that is used alongside with hashing, especially in password storage.



End-to-end encryption

Asymmetric cryptography is used in end-to-end encryption to securely exchange keys, allowing the sender to encrypt data with the recipient's public key while only the recipient can decrypt it with their private key

- Each parties involved in communication will have a public-private key pair, where only public key is share to other.



Blockchain

- it is a distributed record of transaction, gained after computer solving a very difficult math problem
- It uses cryptography in proof of work.
- In proof of work, blockchain uses hash to allow other to verify whether the math problem is solved or not.



Transport Layer Security

- Transport Layer Security is a cryptographic protocol that secures data transmitted over the internet, providing privacy, authentication, and data integrity between applications.
- The way it establish a secure communication channel as follows:
 1. Using 3 way handshake process involving asymmetric encryption to verify and opens a communication channel
 2. Using symmetric encryption to further encrypt data for each communication
- The authentication process is using the public key infrastructure, which is a certificated authority give out certificate which blind in the entity's public key. Allowing both client and server knows each other can be trusted.